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SOCI 20112 1 (Spring 2026): Applications of Hierarchical Linear Models

15 April 2026

Due April 16, 2026 at 12:00PM

Part A: Two-level Cross-sectional Modeling

1. Obtain a point estimate and standard error for the gap using ordinary least squares regression [you can use the HLM software or any stat package for this part]. Call this the “national gap.”

```
hsb1 <- read_sav("C:/Users/theis/Downloads/hsb1.sav")
```

```
model1 <- lm(MATHACH ~ MINORITY, data = hsb1)
summary(model1)
```

Call:

```
lm(formula = MATHACH ~ MINORITY, data = hsb1)
```

Residuals:

Min	1Q	Median	3Q	Max
-16.7144	-5.1309	0.3236	5.3351	15.2401

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.8824	0.0918	151.22	<2e-16 ***
MINORITY	-4.1295	0.1752	-23.58	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.627 on 7183 degrees of freedom

Multiple R-squared: 0.07183, Adjusted R-squared: 0.0717

F-statistic: 555.9 on 1 and 7183 DF, p-value: $< 2.2e-16$

To calculate this regression, I uploaded the hsb1.sav into RStudio. The OLS regression produced a “national gap” estimate of -4.1295 . The estimate has a standard error of 0.1752 . This means that, on average, minority students are expected to score 4.1295 units lower in math achievement compared to majority students. With a p_{value} of $< 2e-16$ and an $\alpha = 0.05$, I conclude that there is statistically significant evidence of a “national gap.”

- 2. Now estimate the average difference (and standard error) between a minority and a majority student that attend the same school using HLM. Call this the average “within-school gap.” (Hint: centering of the minority indicator correctly is essential to get the right estimate.)**

MINORITY has been centered around the group mean.

Mixed Model

$$MATHACH_{ij} = \gamma_{00} + \gamma_{10} * MINORITY_{ij} + u_{0j} + r_{ij}$$

Final Results - Iteration 4

Iterations stopped due to small change in likelihood function

$$\sigma^2 = 37.77494$$

τ
INTRCPT1, β_0 8.65168

Random level-1 coefficient	Reliability estimate
INTRCPT1, β_0	0.905

The value of the log-likelihood function at iteration 4 = -2.343216E+004

Final estimation of fixed effects:

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	12.636436	0.244470	51.689	159	<0.001
For MINORITY slope, β_1					
INTRCPT2, γ_{10}	-3.553638	0.221959	-16.010	7024	<0.001

Final estimation of fixed effects (with robust standard errors)

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	12.636436	0.243688	51.855	159	<0.001
For MINORITY slope, β_1					
INTRCPT2, γ_{10}	-3.553638	0.296233	-11.996	7024	<0.001

Final estimation of variance components

Random Effect	Standard Deviation	Variance Component	d.f.	χ^2	p-value
INTRCPT1, u_0	2.94137	8.65168	159	1720.61548	<0.001
level-1, r	6.14613	37.77494			

Statistics for current covariance components model

Deviance = 46864.328750
Number of estimated parameters = 2

The average “within-school gap” estimated by the model is $\gamma_{10} = -3.55368$. The estimate has a standard error of 0.296233 (or, without robust standard errors, 0.221959). This

means that, when comparing minority and majority students who attend the same school, minority students are expected to score about 3.553628 units lower in math achievement relative to majority students. With a p_{value} of <0.001 and an $\alpha = 0.05$, I reject H_0 and conclude that there is statistically significant evidence of a “within-school gap.”

3. Compare the size of the two gaps estimated in (1) and (2). Provide a substantive explanation for why they are different.

The “national gap” estimate in A.1 was -4.1295 . The “within-school gap” estimated in A.2 was -3.55368 . The “national gap” estimate is ~ 0.56 units larger than the “within-school gap” because the estimate includes both “within-school” and “between-school” differences. In plain English, the “national gap” considers the possibility that minority students may be more likely to attend lower-achieving schools, and vis versa for majority students. Thus, once I remove the “between-school” differences by estimating only the “within-school gap,” some of the potential interference is removed, and the estimated gap becomes smaller. In even *plainer* English, the “national gap” is a *much* broader estimate that reflects student-level and school-level inequality, while the “within-school gap” is narrower because it only compares students attending the same school (as opposed to students across the country attending a wide variety of schools).

4. Consider the extent to which the minority-majority gaps vary from school to school.

4(A): Provide statistical evidence on whether they vary.

Final Results - Iteration 16

Iterations stopped due to small change in likelihood function

$$\sigma^2 = 37.36333$$

$$\tau$$

INTRCPT1, β_0	8.66810	1.70119
MINORITY, β_1	1.70119	3.86548

$$\tau \text{ (as correlations)}$$

INTRCPT1, β_0	1.000	0.294
MINORITY, β_1	0.294	1.000

Random level-1 coefficient	Reliability estimate
INTRCPT1, β_0	0.907
MINORITY, β_1	0.331

Note: The reliability estimates reported above are based on only 136 of 160 units that had sufficient data for computation. Fixed effects and variance components are based on all the data.

The value of the log-likelihood function at iteration 16 = -2.342092E+004

Final estimation of fixed effects:

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	12.633151	0.244539	51.661	159	<0.001
For MINORITY slope, β_1					
INTRCPT2, γ_{10}	-3.501823	0.291700	-12.005	159	<0.001

Final estimation of fixed effects (with robust standard errors)

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	12.633151	0.243773	51.823	159	<0.001
For MINORITY slope, β_1					
INTRCPT2, γ_{10}	-3.501823	0.290441	-12.057	159	<0.001

Final estimation of variance components

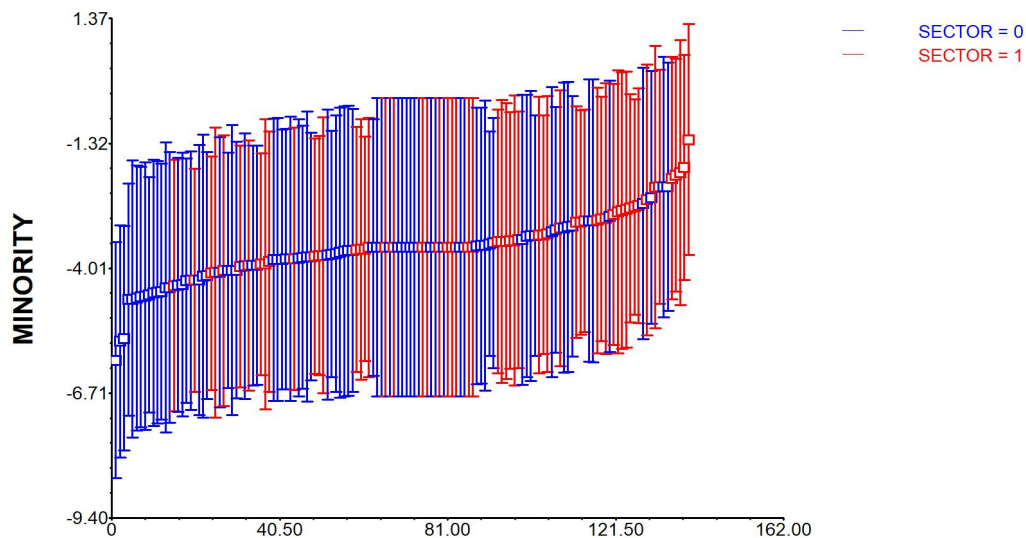
Random Effect	Standard Deviation	Variance Component	d.f.	χ^2	p-value
INTRCPT1, u_0	2.94416	8.66810	135	1410.77800	<0.001
MINORITY slope, u_1	1.96608	3.86548	135	216.68874	<0.001
level-1, r	6.11256	37.36333			

Minority–majority gaps *do* vary across schools. In the “Final estimation of variance components” table, the “Variance Component” for the minority binary variable is 3.86548. What this number tells us is that some schools have *much* larger minority–majority gaps relative to others. With a p_{value} of <0.001 and an $\alpha = 0.05$, I reject H_0 that the variance of the minority slope is 0. Therefore, I conclude that there is statistically significant evidence that the size of the minority–majority gap varies from school to school.

4(B): Do schools with higher achievement tend to have higher or lower minority–majority gaps? Provide statistical evidence (you do not need a hypothesis test).

The Iteration 16 results show us that schools with *higher* math achievement tend to have *smaller* minority–majority gaps. The covariance between the school intercept and minority estimate is 1.70119; the correlation is 0.294. The covariance estimate suggests that schools with higher math achievement, on average, tend to have binary minority estimates closer to 0. In plain English, higher-achieving schools tend to have smaller minority–majority gaps.

4(C): Provide a graph of the empirical Bayes estimates of the gaps, rank ordered in size and color coded by sector. Is there visual evidence that the size of the gaps varies by sector? Explain.



The empirical Bayes graph provides *some*, but not *complete*, evidence that the size of minority–majority gap varies by sector. Sector 0 (the blue CIs) represent public schools while Sector 1 (the red CIs) represent Catholic schools. Visually, it can be argued that there is a much greater concentration of public schools among the more negative minority–majority gap estimates, while the opposite is true for Catholic schools; this *could* suggest that Catholic schools have smaller minority–majority gaps compared to public schools. However, the

re is a pretty significant overlap between the two types of schools, with red and blue confidence intervals appearing across the range of estimates. This, coupled with the, what can be argued as, large confidence intervals suggests that the graph is *suggestive* of Catholic schools having smaller gaps, but certainly not *conclusive*.

- 5. Add school sector to your model at level 2 for both the intercept and the minority gap coefficient. Provide an estimate of the average minority gap in public schools and in Catholic schools. Provide a hypothesis test that these gaps are the same.**

Mixed Model

$$\begin{aligned} MATHACH_{ij} = & \gamma_{00} + \gamma_{01} * SECTOR_{ij} \\ & + \gamma_{10} * MINORITY_{ij} + \gamma_{11} * SECTOR_{ij} * MINORITY_{ij} \\ & + u_{0j} + u_{1j} * MINORITY_{ij} + r_{ij} \end{aligned}$$

Final Results - Iteration 21

Iterations stopped due to small change in likelihood function

$$\sigma^2 = 37.37601$$

τ			
INTRCPT1, β_0	6.72084	-0.89699	
MINORITY, β_1	-0.89699	1.57978	

τ (as correlations)			
INTRCPT1, β_0	1.000	-0.275	
MINORITY, β_1	-0.275	1.000	

Random level-1 coefficient	Reliability estimate
INTRCPT1, β_0	0.883
MINORITY, β_1	0.179

Note: The reliability estimates reported above are based on only 136 of 160 units that had sufficient data for computation. Fixed effects and variance components are based on all the data.
The value of the log-likelihood function at iteration 21 = -2.338846E+004

Final estimation of fixed effects:

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	11.394125	0.292778	38.917	158	<0.001
SECTOR, γ_{01}	2.805383	0.439137	6.388	158	<0.001
For MINORITY slope, β_1					
INTRCPT2, γ_{10}	-5.107432	0.358194	-14.259	158	<0.001
SECTOR, γ_{11}	2.994540	0.507470	5.901	158	<0.001

Final estimation of fixed effects (with robust standard errors)

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	11.394125	0.292332	38.977	158	<0.001
SECTOR, γ_{01}	2.805383	0.435867	6.436	158	<0.001
For MINORITY slope, β_1					
INTRCPT2, γ_{10}	-5.107432	0.362644	-14.084	158	<0.001
SECTOR, γ_{11}	2.994540	0.500604	5.982	158	<0.001

Final estimation of variance components

Random Effect	Standard Deviation	Variance Component	d.f.	χ^2	p-value
INTRCPT1, u_0	2.59246	6.72084	134	1149.42215	<0.001
MINORITY slope, u_1	1.25689	1.57978	134	171.38502	0.016
level-1, r	6.11359	37.37601			

Note: The chi-square statistics reported above are based on only 136 of 160 units that had sufficient data for computation. Fixed effects and variance components are based on all the data.

Statistics for current covariance components model

Deviance = 46776.926517
Number of estimated parameters = 4

In public schools, the average minority gap is given by the minority slope intercept (as mentioned in A.4, public schools are coded as 0). Thus, the average minority gap in public schools is -5.107432 . This estimate suggests that, on average, minority students in public schools are expected to score ~ 5.11 units lower on math achievement relative to majority students in public schools.

For Catholic schools, the average minority gap is $-5.107432 + 2.994540 = -2.112892$. This estimate suggests that, on average, minority students in Catholic schools are expected to score ~ 2.11 units lower on math achievement relative to majority students in Catholic schools.

To test my hypothesis, I will specifically look at the p_{value} of the γ_{11} estimate. With a p_{value} of <0.001 and an $\alpha = 0.05$, I reject H_0 that there is no difference in the minority–majority gap between public and Catholic schools. Therefore, I conclude that the minority–majority gap is statistically significantly different between public and Catholic schools.

- It is reasonable to suggest that the minority gaps might depend on the racial makeup of the school. The indicator variable HIMINTY is coded 1 if minority enrollment in a school exceeds 40%. Add this indicator to your model along with sector for both level 1 terms.

Mixed Model

$$\begin{aligned} MATHACH_{ij} = & \gamma_{00} + \gamma_{01} * SECTOR_j + \gamma_{02} * HIMINTY_j \\ & + \gamma_{10} * MINORITY_{ij} + \gamma_{11} * SECTOR_j * MINORITY_{ij} + \gamma_{12} * HIMINTY_j * MINORITY_{ij} \\ & + u_{0j} + u_{1j} * MINORITY_{ij} + r_{ij} \end{aligned}$$

Final Results - Iteration 21

Iterations stopped due to small change in likelihood function

$$\sigma^2 = 37.38499$$

τ

INTRCPT1, β_0	5.11077	-0.73264
MINORITY, β_1	-0.73264	1.62373

τ (as correlations)

INTRCPT1, β_0	1.000	-0.254
MINORITY, β_1	-0.254	1.000

Random level-1 coefficient	Reliability estimate
INTRCPT1, β_0	0.852
MINORITY, β_1	0.183

Note: The reliability estimates reported above are based on only 136 of 160 units that had sufficient data for computation. Fixed effects and variance components are based on all the data.

The value of the log-likelihood function at iteration 21 = -2.336966E+004

Final estimation of fixed effects:

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. df.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	12.109481	0.282994	42.791	157	<0.001
SECTOR, γ_{01}	2.930040	0.390227	7.509	157	<0.001
HIMINTY, γ_{02}	-2.801386	0.434256	-6.451	157	<0.001
For MINORITY slope, β_1					
INTRCPT2, γ_{10}	-5.162064	0.405107	-12.742	157	<0.001
SECTOR, γ_{11}	2.987175	0.510607	5.850	157	<0.001
HIMINTY, γ_{12}	0.249355	0.523523	0.476	157	0.635

**Final estimation of fixed effects
(with robust standard errors)**

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	12.109481	0.285590	42.402	157	<0.001
SECTOR, γ_{01}	2.930040	0.382070	7.669	157	<0.001
HIMINTY, γ_{02}	-2.801386	0.462455	-6.058	157	<0.001
For MINORITY slope, β_1					
INTRCPT2, γ_{10}	-5.162064	0.358130	-14.414	157	<0.001
SECTOR, γ_{11}	2.987175	0.514285	5.808	157	<0.001
HIMINTY, γ_{12}	0.249355	0.532279	0.468	157	0.640

Final estimation of variance components

Random Effect	Standard Deviation	Variance Component	d.f.	χ^2	p-value
INTRCPT1, u_0	2.26070	5.11077	133	941.33026	<0.001
MINORITY slope, u_1	1.27426	1.62373	133	171.51714	0.014
level-1, r	6.11433	37.38499			

Note: The chi-square statistics reported above are based on only 136 of 160 units that had sufficient data for computation. Fixed effects and variance components are based on all the data.

Statistics for current covariance components model

Deviance = 46739.316331
Number of estimated parameters = 4

6(A): Interpret the coefficients for the new HIMINITY terms reported by your model

1. The HIMINTY, γ_{02} estimate is -2.801386. This estimate shows us the difference in average math achievement scores between schools where minority enrollment *exceeds* 40%, and schools where minority enrollment *does not* exceed 40%. This means that, holding constant school sector, I expect that schools with minority enrollment above 40% will have math achievement scores that are ~2.80 units low than schools with minority enrollment below 40%. With a p_{value} of <0.001 and an $\alpha = 0.05$, I reject H_0 that $\gamma_{02} = 0$.
2. The HIMINTY, γ_{12} estimate is 0.249355. This estimate means that in schools where minority enrollment is *above* 40%, the minority-majority gap is predicted to be 0.25 points smaller than in schools where minority enrollment is below 40%, holding constant school sector. However, with a p_{value} of 0.640 and an $\alpha = 0.05$, I fail reject H_0 that $\gamma_{12} = 0$ and cannot conclude that the minority-majority gap differs by school minority composition.

6(B): How does adding racial makeup to the model change our understanding of how school sector affects math achievement? Explain

Before adding the racial makeup to the model, the $\gamma_{01} = 2.805383$. This estimate shows us the difference in average math achievement between Catholic and public schools. In other words, Catholic schools were predicted to score ~2.81 units higher than public schools on average. However, after adding the racial makeup to the model, the coefficient increased to 2.930040. This means that Catholic schools still had higher average math achievement even after controlling for whether minority enrollment exceeded 40%. The $\gamma_{11} = 2.994540$ estimate increased to 2.987175 after adding racial makeup to the model. This estimate suggests that the minority–majority gap is ~2.99 units less negative in Catholic schools compared to public schools. If anything, adding the racial makeup only confirmed the evidence suggesting that Catholic schools have higher math achievement and smaller minority–majority gaps than public schools, on average, even after including whether the school has a minority enrollment of above 40%.

7: Write a summary paragraph explaining what you have learned about the minority majority math achievement gap in the US in 1980.

The regressions above suggest that there was a clear minority–majority math achievement gap between White students and Black/Hispanic students in 1980. Across the country (e.g. across all schools), minority students scored an average of 4.13 units *lower* than White students, on average. Within the same school, minority students scored an average of 3.57 units lower than White students, on average; this shows that perhaps part of the national gap can be explained by the school a student attends. Furthermore, the data revealed that higher-achieving schools and Catholic schools tend to have smaller minority–majority gaps. Overall, the regressions suggest that, in 1980, the minority–majority math achievement gap varied across schools and was smaller in Catholic schools than public schools; but across all estimates, minority students were expected to score lower than majority students, on average.

Part B. Two-level Growth modeling

- 1: Write down a level-1 model that includes a linear and quadratic growth terms. For this model, age should be centered around 18 months. Define each π and e_{ti} .

$$VOCAB_{ti} = \pi_{0i} + \pi_{1i} * (AGE_{18}_{ti}) + \pi_{2i} * (AGE_{18}SE_{ti}) + e_{ti}$$

π_{0i} shows the predicted vocabulary size for child_i at 18 months.

π_{1i} shows the linear vocabulary growth-rate for child_i at 18 months.

π_{2i} shows the quadratic vocabulary growth-rate for child_i.

e_{ti} shows the Level-1 error term and represents the difference between child_i's *actual* vocabulary at time t and their *predicted* vocabulary from the model.

- 2: Write down a level-2 model in which each level 1 term is allowed to vary randomly. Define each β and each r

$$\pi_{0i} = \beta_{00} + r_{0i}$$

$$\pi_{1i} = \beta_{10} + r_{1i}$$

$$\pi_{2i} = \beta_{20} + r_{2i}$$

β_{00} shows the average vocabulary size across all children at 18 months.

β_{10} shows the average linear growth rate in vocabulary at 18 months across all children.

β_{20} shows the quadratic linear growth rate in vocabulary at 18 months across all children.

r_{0i} shows child_i's deviation from the overall average vocabulary size at 18 months.

r_{1i} shows child_i's deviation from the average linear growth rate in vocabulary at 18 months across all children.

r_{2i} shows child_{*i*}'s deviation from the average quadratic vocabulary growth–rate across all children.

- 3: Run the model using Full Maximum Likelihood estimation. Explain in substantive terms what your estimates for each beta means for the shape of your estimated growth trajectories**

Mixed Model

$$\begin{aligned} VOCAB_{ij} = & \gamma_{00} \\ & + \gamma_{10} * AGE18_{ij} \\ & + \gamma_{20} * AGE18SQ_{ij} + u_{0j} + u_{1j} * AGE18_{ij} + u_{2j} * AGE18SQ_{ij} + r_{ij} \end{aligned}$$

Final Results - Iteration 1314

Iterations stopped due to small change in likelihood function

$$\sigma^2 = 664.16365$$

Standard error of $\sigma^2 = 117.18410$

τ				
INTRCPT1, β_0	3016.99373	808.27860	34.28097	
AGE18, β_1	808.27860	217.80237	9.40902	
AGE18SQ, β_2	34.28097	9.40902	0.43011	

Standard errors of τ				
INTRCPT1, β_0	989.59573	252.00921	12.32786	
AGE18, β_1	252.00921	67.69826	3.43325	
AGE18SQ, β_2	12.32786	3.43325	0.24043	

τ (as correlations)				
INTRCPT1, β_0	1.000	0.997	0.952	
AGE18, β_1	0.997	1.000	0.972	
AGE18SQ, β_2	0.952	0.972	1.000	

Confidence intervals of τ correlations				
INTRCPT1, β_0	(1.000, 1.000)	(-1.000,-1.#IO)	(-1.000,-1.#IO)	
AGE18, β_1	(-1.000,-1.#IO)	(1.000, 1.000)	(-1.000,-1.#IO)	
AGE18SQ, β_2	(-1.000,-1.#IO)	(-1.000,-1.#IO)	(1.000, 1.000)	

Random level-1 coefficient	Reliability estimate
INTRCPT1, β_0	0.908
AGE18, β_1	0.969
AGE18SQ, β_2	0.516

The value of the log-likelihood function at iteration 1314 = -6.387340E+002

Final estimation of fixed effects:

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	67.473747	12.211290	5.526	21	<0.001
For AGE18 slope, β_1					
INTRCPT2, γ_{10}	24.084255	3.194468	7.539	21	<0.001
For AGE18SQ slope, β_2					
INTRCPT2, γ_{20}	2.037055	0.189657	10.741	21	<0.001

**Final estimation of fixed effects
(with robust standard errors)**

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. df.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	67.473747	12.197723	5.532	21	<0.001
For AGE18 slope, β_1					
INTRCPT2, γ_{10}	24.084255	3.166658	7.606	21	<0.001
For AGE18SQ slope, β_2					
INTRCPT2, γ_{20}	2.037055	0.182649	11.153	21	<0.001

Final estimation of variance components

Random Effect	Standard Deviation	Variance Component	df.	χ^2	p-value
INTRCPT1, u_0	54.92717	3016.99373	21	319.32830	<0.001
AGE18 slope, u_1	14.75813	217.80237	21	825.62425	<0.001
AGE18SQ slope, u_2	0.65583	0.43011	21	49.91993	<0.001
level-1, r	25.77137	664.16365			

Statistics for the current model

Deviance = 1277.467938
Number of estimated parameters = 10

The estimate for β_0 is 67.473747. This suggests that the average child is expected to have a vocabulary of ~67 words at 18 months.

$$H_0: \beta_0 = 0$$

$$H_A: \beta_0 \neq 0$$

$$\alpha = 0.05$$

With a p_{value} of < 0.0001 and an $\alpha = 0.05$, I reject H_0 that β_0 equals 0. This means the predicted average vocabulary at 18 months is statistically different from 0.

The estimate for β_1 is 24.084255. This suggests that children are expected to gain about 24.08 words per month.

$$H_0: \beta_1 = 0$$

$$H_A: \beta_1 \neq 0$$

$$\alpha = 0.05$$

With a p_{value} of < 0.0001 and an $\alpha = 0.05$, I reject H_0 that β_1 equals 0. This means the average linear vocabulary growth rate at 18 months is statistically different from 0.

The estimate for β_2 is 2.037055. This suggests a child's vocabulary growth is not only increasing *linearly* over time, but increasing at a faster rate as they get older.

$$H_0: \beta_2 = 0$$

$$H_A: \beta_2 \neq 0$$

$$\alpha = 0.05$$

With a p_{value} of < 0.0001 and an $\alpha = 0.05$, I reject H_0 that β_2 equals 0. This means the average quadratic vocabulary growth is statistically different from 0.

4: Add the measure of words spoken by the mother at the 16-month observation

(MOMSPEAK) to the model at level 2 for each level 1 term. Define β_{00} , β_{01} , β_{10} , and β_{11} .

$$\pi_{0i} = \beta_{00} + \beta_{01}(MOMSPEAK_i) + r_{0i}$$

$$\pi_{1i} = \beta_{10} + \beta_{11}(MOMSPEAK_i) + r_{1i}$$

$$\pi_{2i} = \beta_{20} + \beta_{21}(MOMSPEAK_i) + r_{2i}$$

β_{00} shows the estimated vocabulary size at 18 months for a child whose mother spoke 0 words at the 16-month observation.

β_{01} shows the expected change in vocabulary size at 18 months for each additional word spoken by the mother at the 16-month observation.

β_{10} shows the estimated linear vocabulary growth rate at 18 months for a child whose mother spoke 0 words at the 16-month observation.

β_{11} shows the expected change in the linear vocabulary growth rate at 18 months for each additional word spoken by the mother at the 16-month observation.

5: Run the model with MOMSPEAK and explain in a paragraph what you found

Mixed Model

$$\begin{aligned} VOCAB_{ij} = & \gamma_{00} + \gamma_{01} * MOMSPEAK_j \\ & + \gamma_{10} * AGE18_{ij} + \gamma_{11} * MOMSPEAK_j * AGE18_{ij} \\ & + \gamma_{20} * AGE18SQ_{ij} + \gamma_{21} * MOMSPEAK_j * AGE18SQ_{ij} \\ & + u_{0j} + u_{1j} * AGE18_{ij} + u_{2j} * AGE18SQ_{ij} + r_{ij} \end{aligned}$$

Final Results - Iteration 1793

Iterations stopped due to small change in likelihood function

$$\sigma^2 = 653.84070$$

Standard error of $\sigma^2 = 114.94188$

τ			
INTRCPT1, β_0	2567.96271	655.12103	22.43076
AGE18, β_1	655.12103	167.19578	5.73287
AGE18SQ, β_2	22.43076	5.73287	0.19785

Standard errors of τ			
INTRCPT1, β_0	852.29743	205.19226	9.02678
AGE18, β_1	205.19226	52.40402	2.37060
AGE18SQ, β_2	9.02678	2.37060	0.16749

τ (as correlations)			
INTRCPT1, β_0	1.000	1.000	0.995
AGE18, β_1	1.000	1.000	0.997
AGE18SQ, β_2	0.995	0.997	1.000

Confidence intervals of τ correlations			
INTRCPT1, β_0	(1.000, 1.000)	(-1.000,-1.#IO)	(-1.000,-1.#IO)
AGE18, β_1	(-1.000,-1.#IO)	(1.000, 1.000)	(-1.000,-1.#IO)
AGE18SQ, β_2	(-1.000,-1.#IO)	(-1.000,-1.#IO)	(1.000, 1.000)

Random level-1 coefficient	Reliability estimate
INTRCPT1, β_0	0.895
AGE18, β_1	0.961
AGE18SQ, β_2	0.342

The value of the log-likelihood function at iteration 1793 = -6.341688E+002

Final estimation of fixed effects:

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	23.207187	25.571328	0.908	20	0.375
MOMSPEAK, γ_{01}	0.013130	0.006865	1.913	20	0.070
For AGE18 slope, β_1					
INTRCPT2, γ_{10}	9.743927	6.337227	1.538	20	0.140
MOMSPEAK, γ_{11}	0.004308	0.001702	2.531	20	0.020
For AGE18SQ slope, β_2					
INTRCPT2, γ_{20}	1.036625	0.357321	2.901	20	0.009
MOMSPEAK, γ_{21}	0.000307	0.000096	3.216	20	0.004

**Final estimation of fixed effects
(with robust standard errors)**

Fixed Effect	Coefficient	Standard error	t-ratio	Approx. d.f.	p-value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	23.207187	17.287979	1.342	20	0.195
MOMSPEAK, γ_{01}	0.013130	0.004745	2.767	20	0.012
For AGE18 slope, β_1					
INTRCPT2, γ_{10}	9.743927	4.483501	2.173	20	0.042
MOMSPEAK, γ_{11}	0.004308	0.001258	3.425	20	0.003
For AGE18SQ slope, β_2					
INTRCPT2, γ_{20}	1.036625	0.272010	3.811	20	0.001
MOMSPEAK, γ_{21}	0.000307	0.000080	3.846	20	0.001

Final estimation of variance components

Random Effect	Standard Deviation	Variance Component	d.f.	χ^2	p-value
INTRCPT1, u_0	50.67507	2567.96271	20	270.98442	<0.001
AGE18 slope, u_1	12.93042	167.19578	20	648.66942	<0.001
AGE18SQ slope, u_2	0.44480	0.19785	20	31.81035	0.045
level-1, r	25.57031	653.84070			

Statistics for the current model

Deviance = 1268.337550

Number of estimated parameters = 13

The regression suggests that the number of words spoken by the mother at the 16-month observation is positively associated with their child's vocabulary. The γ_{01} estimate is 0.013130; this estimate suggests that children whose mothers spoke words at the 16-month observation were expected to have a larger vocabulary at 18-months (specifically: $23.207187 + 0.013130 = 23.220317$ words). The γ_{11} estimate is 0.004308. This estimates that the number of words spoken by the mother at the 16-month observation is associated with a faster vocabulary growth-rate at 18 months. The γ_{21} estimate is 0.000307. This estimate suggests that the number of words spoken by the mother at the 16-month observation is associated with a faster vocab growth-rate over time. Together, the findings suggest that children

whose mothers spoke more words at 16 months are expected to have higher vocabulary levels and faster vocabulary growth at 18 months, on average.

- 6: Create a model graph to visualize the results of the final model, showing the model-based trajectories for children whose mothers are at the 25th and 75th percentiles of MOMSPEAK.**

